Expernetic(Pvt) Ltd

Assignment
K.P.Sithmi Maheema Jayawan

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Flexibility to work onsite - **Yes, I am available to work onsite and can adapt to the company's work schedule and location.**

1 Executive Summary A 1-page business-oriented summary of your most important findings and recommendations
2 Objectives & Scope What you set out to achieve, the ci es selected, and your prioritization rationale
3 Dataset Overview Dataset description, files used, relationships, limitations, and assumptions
4 Methodology Your overall analytical approach and the reasoning behind key method selections
5 Engineering Approach Pipeline design, architecture decisions, data model, and production considerations
6 EDA Findings Key pa erns, visualizations, and their business interpretations
7 Statistical Findings Hypothesis results, effect sizes, and statistical conclusions in business terms
8 Data Science Experiments Model results, comparison tables, validation metrics, and interpretation
9 AI/ML Experiments NLP, LLM, or agentic work, including methodology, results, and critical evaluation.
10 Visualizations High-quality figures with numbered captions and clear interpretations
11 Business Recommendations Actionable recommendations derived from your analysis
12 Cross-City Comparisons If applicable: comparative insights, regional pa erns, and market dynamics
13 Limitations & Caveats Honest discussion of data limitations, model weaknesses, and scope boundaries
14 Future Improvements What you would do next with more me, be er data, or additional resources

1.Executive Summary

This project analyzes the Airbnb marketplace in **Boston** to identify the key factors influencing listing prices, customer satisfaction, availability, and neighborhood-level market behavior. The objective was to transform raw Airbnb data into meaningful business insights that can support hosts and marketplace stakeholders in making data-driven pricing and operational decisions.

A complete data analytics pipeline was developed by integrating multiple Airbnb datasets, including listing information, calendar availability, reviews, and neighborhood details. The data processing workflow included data exploration, cleaning, validation, feature engineering, and dataset integration to create a reliable analytical dataset. Due to computational limitations, a single-city analysis approach was selected to prioritize depth, data quality, and analytical accuracy over broader but less detailed comparisons. This approach aligns with the principle that high-quality analysis of one market provides more valuable insights than superficial analysis across multiple markets.

The exploratory analysis revealed that Airbnb pricing is strongly influenced by property characteristics, especially accommodation capacity, number of bedrooms, room type, and location. Entire-home listings generally command higher prices compared with private rooms due to increased privacy, space, and guest capacity. Neighborhood-level analysis showed noticeable price variation across different areas, highlighting the importance of location as a major factor in marketplace pricing strategies.

Statistical analysis was performed to validate key business assumptions through hypothesis testing, confidence interval analysis, effect size measurement, correlation analysis, and regression modelling. The regression analysis identified the major drivers of listing prices while controlling for multiple factors simultaneously. Multicollinearity testing using Variance Inflation Factor (VIF) confirmed that the selected predictors were reliable and suitable for interpretation. The findings indicate that property size, guest capacity, and listing characteristics have a measurable impact on pricing decisions.

Power BI dashboards were developed to provide interactive business insights through KPI cards, slicers, matrix analysis, drill-down reports, and drill-through functionality. These dashboards enable users to explore listing performance, pricing patterns, availability trends, and neighborhood differences from different analytical perspectives.

Based on the analysis, several business recommendations were identified:

1. **Implement data-driven pricing strategies**
   Hosts should consider property characteristics, neighborhood demand, and seasonal availability when setting prices rather than relying on fixed pricing approaches.
2. **Optimize listing features and capacity**
   Properties with greater accommodation capacity and additional facilities have stronger pricing potential. Hosts should highlight valuable amenities and improve listing quality to increase competitiveness.
3. **Use neighborhood-based market insights**
   Pricing strategies should be customized according to local market conditions because different neighborhoods demonstrate different demand and pricing behaviors.
4. **Improve availability management**
   Monitoring occupancy and availability patterns can help hosts identify demand opportunities and adjust pricing strategies during high-demand periods.

5. **Develop predictive analytics solutions**
   Future improvements could include machine learning models for automated price prediction, demand forecasting, and personalized recommendations to support smarter marketplace decisions.

Overall, this project demonstrates the application of data engineering, statistical analysis, business intelligence, and data science techniques to solve a real-world marketplace analytics problem. The developed analytical framework provides actionable insights that can help Airbnb stakeholders improve pricing decisions, increase revenue opportunities, and better understand customer and market behavior.

2.Objectives & Scope

2.1 Objectives

The main objective of this project is to analyze the **Boston Airbnb marketplace** and generate data-driven insights into pricing patterns, listing performance, customer reviews, and market behavior. The project aims to:

- Identify the key factors influencing Airbnb listing prices.
- Analyze differences in pricing across room types and neighborhoods.
- Evaluate the relationship between host performance, customer ratings, and listing characteristics.
- Develop statistical models to understand major price drivers.
- Create interactive dashboards to support business decision-making.
- Provide recommendations to help hosts improve pricing strategies and marketplace competitiveness.

2.2 Scope

This project focuses on the analysis of Airbnb data for **Boston** using multiple datasets, including:

- **Listings data** – property details, host information, room types, amenities, and prices.
- **Calendar data** – availability patterns and occupancy-related information.
- **Reviews data** – customer feedback and rating information.
- **Neighborhood data** – location-based market analysis.

The project covers the complete analytics workflow, including data exploration, data cleaning, feature engineering, statistical analysis, visualization, and business intelligence reporting using Power BI.

2.3 City Selection & Prioritization Rationale

Initially, multiple Airbnb city datasets were considered. However, due to hardware limitations, including **8 GB RAM and limited system storage availability**, a single-city analysis approach was selected to ensure efficient processing and maintain analytical quality.

Boston was selected because it provides a suitable balance between dataset size and analytical value. The dataset contains sufficient listing, review, calendar, and neighborhood information to perform meaningful statistical analysis while remaining manageable for local processing.

Prioritizing one city allowed deeper analysis of market trends, pricing behavior, and business insights rather than performing a broader analysis with reduced accuracy and limited depth. This approach ensured reliable results while maintaining computational efficiency.

3.Dataset Overview

3.1 Dataset Description

This project uses the Boston Airbnb Open Data dataset to analyze marketplace behaviour, pricing patterns, availability trends, and customer review performance. The dataset contains information about Airbnb listings, hosts, calendar availability, guest reviews, and neighbourhood details.

The dataset was selected because it provides sufficient information to perform data cleaning, feature engineering, statistical analysis, data modelling, and business intelligence reporting while remaining manageable for local processing.

The raw datasets were processed through a data pipeline involving:

- Data exploration
- Data cleaning and standardization
- Feature engineering
- Data validation
- Dataset integration
- SQL-based data modelling
- Power BI visualization

The final output was an analytical master dataset and a star schema data warehouse model.

Figure 1 Star Schema

3.2 Files Used

The following datasets were used in the analysis:

Dataset	Description	Key Information
listings.csv	Contains property and host details	Listing ID, host information, room type, property type, price, amenities, ratings
calendar.csv	Contains daily availability information	Listing ID, date, availability status, price, minimum nights
reviews.csv	Contains guest review information	Listing ID, review date, reviewer details, comments
neighbourhoods.csv	Contains location information	Neighbourhood names and geographic grouping details

After data cleaning and feature engineering, these datasets were combined into a master dataset for analysis.

3.3 Dataset Relationships

The datasets are connected through common identifiers.

3.4 Primary and Foreign Keys

Dataset	Key	Type	Description
Listings	id	Primary Key	Unique identifier for each Airbnb listing
Calendar	listing_id	Foreign Key	References Airbnb listings
Reviews	listing_id	Foreign Key	References Airbnb listings
Neighbourhoods	neighbourhood	Reference Attribute	Used for location-based analysis

3.5 Dataset Limitations

Several limitations were identified during the analysis:

- The dataset represents Airbnb activity at a specific period and may not reflect current market conditions.
- Some fields contain missing values due to incomplete host or guest information.
- Calendar availability does not always represent confirmed bookings because blocked dates may include owner reservations or unavailable periods.
- Review data depends on guest participation, meaning some listings may have limited feedback.
- Pricing can change due to seasonal demand, special events, and market conditions.
- Neighbourhood information is limited to the classifications provided within the dataset.

3.6 Assumptions

The following assumptions were made during data preparation and analysis:

- The listing ID uniquely identifies each Airbnb property.
- Price values represent the daily rental price of a listing.
- Availability information is used as an indicator of potential occupancy behaviour.

- Review scores are considered a measure of customer satisfaction.
- Missing values were handled using appropriate cleaning strategies without significantly affecting analysis results.
- Neighbourhood information is assumed to influence pricing and demand patterns.
- Duplicate records identified during validation were removed to maintain data accuracy.

3.7 Business Domain Context

The Airbnb marketplace consists of several important entities:

Listing Entity:
Represents individual Airbnb properties available for rental. It includes property characteristics, pricing information, room type, amenities, and availability.

Host Entity:
Represents individuals or organizations managing Airbnb properties. Host attributes such as Superhost status and verification status help evaluate host performance.

Review Entity:
Represents customer feedback after stays and provides insights into guest satisfaction and service quality.

Calendar Entity:
Represents daily availability and pricing changes, allowing analysis of occupancy and revenue patterns.

Neighbourhood Entity:
Represents geographical areas used to analyze location-based pricing differences and market trends.
This dataset structure enables analysis of Airbnb performance, pricing drivers, customer behaviour, and location-based market opportunities.

4.Methodology

This project follows an end-to-end **data analytics methodology** to transform raw Airbnb data into meaningful business insights. The approach combines data engineering, statistical analysis, business intelligence, and data modelling techniques to understand pricing behaviour, listing performance, and market trends.

The overall workflow consists of the following stages:

4.1 Data Collection and Ingestion

The analysis began by collecting the **Boston Airbnb dataset**, which includes listing details, calendar availability, reviews, and neighbourhood information. The datasets were imported and inspected to understand their structure, data types, relationships, and quality issues.
A single-city approach was selected because Boston provides sufficient data volume for meaningful analysis while allowing efficient processing and detailed investigation of marketplace behaviour.

4.2 Data Exploration and Profiling

Initial exploratory analysis was performed to understand dataset characteristics and identify potential data quality issues.
The profiling process included:

- Row and column count analysis
- Data type inspection
- Missing value analysis
- Duplicate record detection
- Unique value examination
- Distribution analysis of important numerical variables

This step helped identify inconsistencies and determine appropriate cleaning strategies before analysis.

4.3 Data Cleaning and Standardization

Python with **Pandas** was used for data cleaning and preparation.
The following activities were performed:

- Standardizing price fields by removing currency symbols and converting values into numeric format
- Formatting date fields consistently
- Handling missing values using appropriate strategies
- Removing duplicate records
- Standardizing categorical variables such as room type and property type
- Validating numerical fields including price, availability, and location coordinates

The cleaned datasets were then prepared for further analysis and integration.

4.4 Feature Engineering and Data Enrichment

Additional analytical features were created to improve business insights.
The following features were generated:

- Occupancy rate based on calendar availability
- Estimated revenue metrics
- Host tenure on the platform
- Review frequency
- Price per bedroom
- Neighbourhood-level price and rating summaries

Calendar and review information were aggregated and combined with listing data to create a comprehensive master dataset.

4.5 Data Modelling

A **Star Schema data warehouse model** was designed to support efficient analysis and Power BI reporting. The model consists of:

- **Fact_Airbnb** – Stores measurable metrics such as price, availability, occupancy rate, and review scores.
- **Dim_Property** – Contains property-related attributes.
- **Dim_Host** – Contains host information.
- **Dim_Neighbourhood** – Stores geographical information.
- **Dim_Date** – Supports time-based analysis.

A star schema was selected because it provides:

- Improved query performance
- Simplified reporting
- Better compatibility with Power BI
- Clear separation between measures and descriptive attributes

4.6 Statistical Analysis Using R

R was used to perform statistical modelling and validate business assumptions about the Airbnb marketplace.

The following statistical techniques were applied:

- Hypothesis testing using t-tests and ANOVA to evaluate differences between groups.
- Confidence interval analysis to estimate average price differences.
- Effect size calculations to measure practical significance.
- Correlation analysis to identify relationships between numerical variables.
- Regression modelling (OLS) to identify key drivers influencing listing prices.
- Variance Inflation Factor (VIF) analysis to check multicollinearity among predictors.

R was selected because it provides strong statistical libraries and tools for hypothesis testing, modelling, and statistical interpretation.

4.7 Business Intelligence and Visualization

Power BI was used to develop interactive dashboards for business users.
The dashboards include:

- KPI cards for marketplace performance indicators

- Slicers for interactive filtering
- Matrix visualizations for detailed analysis
- Drill-down reports for hierarchical exploration
- Drill-through pages for detailed listing-level insights

These visualizations allow users to explore pricing trends, availability patterns, and neighbourhood performance.

4.8 Method Selection Rationale

The selected tools were chosen based on their strengths and suitability for different stages of the analytical workflow:

- Python (Jupyter Notebook with Pandas) was used for data ingestion, exploration, cleaning, transformation, and feature engineering. Jupyter Notebook provided an interactive environment for executing code, documenting the analysis process, and validating data preparation steps.
- SQL Server was used for structured data storage, analytical querying, and implementation of the star schema data warehouse model. It provides an efficient environment for managing the processed Airbnb dataset and supporting business intelligence reporting.
- R was used for statistical analysis because it provides strong statistical packages for hypothesis testing, confidence interval analysis, correlation analysis, regression modelling, effect size calculation, and multicollinearity testing.
- Power BI was used for visualization and reporting because it enables interactive dashboards, KPI analysis, slicers, drill-down reports, and drill-through analysis to communicate insights effectively to business users.

This combination of Jupyter Notebook (Python), SQL Server, R, and Power BI provides a complete analytical workflow from raw data processing to statistical validation and business decision support.

5.Engineering Approach

The engineering approach focuses on designing a reliable data pipeline that transforms raw Airbnb data into clean, validated, and analytics-ready datasets. The pipeline was designed with scalability, data quality, and analytical performance in mind.

5.1 Pipeline Design

The implemented pipeline follows a structured **Extract–Transform–Load (ETL)** approach:

- **Extract:**
  Raw Airbnb datasets (listings, calendar, reviews, and neighbourhood files) were collected and loaded into the processing environment.
- **Transform:**
  Data cleaning and transformation steps were performed using **Python in Jupyter Notebook**. This included handling missing values, removing duplicates, standardizing formats, validating data types, and creating new analytical features.
- **Load:**
  The processed datasets were loaded into **SQL Server**, where a dimensional data warehouse model was created for reporting and analysis.

5.2 Architecture Decisions

The following architecture decisions were made:

- A **single-city analysis approach** was selected using the Boston Airbnb dataset to maintain data quality and allow detailed analysis while considering available computing resources.
- A modular pipeline structure was followed, separating different processing stages:
  - Data ingestion
  - Data cleaning
  - Feature engineering
  - Validation
  - Database loading
- Python was selected for data processing because it provides efficient libraries such as Pandas for handling structured datasets.
- SQL Server was selected as the storage layer because it supports relational data management, SQL querying, and integration with Power BI.
- Power BI was selected as the visualization layer because it provides interactive reporting capabilities for business users.

5.3 Data Model Design

A **Star Schema** data model was implemented to support efficient analytical queries and reporting.
The model consists of:

Fact Table

Fact_Airbnb

Contains measurable business metrics:

- Listing price
- Availability metrics
- Occupancy rate

- Revenue estimates
- Review counts
- Review scores

Dimension Tables

Dim_Property

- Property type
- Room type
- Number of bedrooms
- Accommodation capacity
- Amenities

Dim_Host

- Host information
- Superhost status
- Verification status
- Host listing count

Dim_Neighbourhood

- Neighbourhood name
- Location information

Dim_Date

- Date
- Year
- Month
- Quarter
- Day information

The star schema provides a simplified structure where dimension tables describe entities and the fact table stores analytical measurements.

5.4 Data Quality and Validation

Data quality checks were implemented throughout the pipeline to improve reliability.
The validation process included:

- Checking missing values
- Identifying duplicate records
- Validating numerical ranges
- Detecting invalid prices
- Checking geographic coordinate validity
- Ensuring consistent data types
- Confirming relationships between datasets

Invalid records were either removed or handled using appropriate cleaning strategies.

5.5 Production Considerations

Although this project was developed as an analytical solution, several production considerations were considered:

- **Automation:**
  The pipeline structure allows future automation by scheduling ingestion and transformation processes.
- **Scalability:**
  The modular design allows additional Airbnb cities to be processed with minimal changes.
- **Error Handling:**
  Validation steps help identify data quality issues before loading data into the warehouse.
- **Incremental Processing:**
  Future improvements could include incremental loading to process only newly added or updated records instead of reprocessing the complete dataset.
- **Metadata and Lineage Tracking:**
  Dataset processing stages can be tracked to maintain transparency from raw data sources to final analytical outputs.
- **Security and Reliability:**
  Database access controls and structured data storage improve data management and reliability.

Overall, the engineering approach provides a complete data pipeline from raw Airbnb data ingestion to analytics-ready datasets. The combination of Python, SQL Server, R, and Power BI creates a reliable foundation for statistical analysis, reporting, and future machine learning applications.

6.EDA Findings

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the Boston Airbnb marketplace and identify patterns related to pricing, availability, reviews, and host performance. The analysis provided valuable insights that guided the subsequent statistical modelling and business recommendations.

6.1 Listing Distribution

The Boston dataset contains thousands of Airbnb listings distributed across multiple neighbourhoods and property types. The majority of listings are concentrated in a small number of popular neighbourhoods, indicating higher tourism and accommodation demand in these areas.

Business Interpretation

- High listing concentration suggests strong market demand in specific neighbourhoods.
- New hosts may achieve better visibility by targeting less saturated locations.
- Location remains one of the most important factors influencing listing performance.

6.2 Price Distribution

The distribution of listing prices was highly right-skewed, with most listings concentrated within lower and mid-price ranges and a small number of luxury properties charging substantially higher prices.

Business Interpretation

- The Airbnb market is primarily driven by affordable and mid-range accommodations.
- Luxury listings represent a niche market segment.
- Median prices provide a more reliable indicator of market pricing than average prices due to extreme outliers.

6.3 Room Type Analysis

Entire-home listings generally exhibited higher prices compared with private and shared rooms.

Business Interpretation

- Guests are willing to pay a premium for privacy and additional space.
- Entire-home listings offer greater revenue potential for hosts.
- Room type should be considered when developing pricing strategies.

6.4 Neighbourhood-Level Pricing Patterns

Significant variation in average listing prices was observed across neighbourhoods.

Business Interpretation

- Location strongly influences pricing decisions.
- Premium neighbourhoods can command higher nightly rates.
- Hosts should benchmark prices against comparable listings within the same neighbourhood.

6.5 Availability and Occupancy Trends

Listings with lower availability generally indicate higher occupancy levels and stronger demand.

Business Interpretation

- High occupancy rates may reflect desirable properties or locations.
- Monitoring availability patterns can help hosts optimize pricing during peak demand periods.
- Availability data can be used to estimate potential revenue opportunities.

6.6 Review Activity and Ratings
Most listings received positive review ratings, while review volume varied considerably across properties.
Business Interpretation
- Strong review ratings contribute to guest trust and booking confidence.
- Listings with more reviews tend to have greater visibility and credibility.
- Maintaining service quality is important for long-term marketplace success.

6.7 Superhost Performance
Superhosts generally demonstrated stronger performance in terms of review quality and guest satisfaction.
Business Interpretation
- Superhost status can serve as a quality signal for potential guests.
- Hosts may benefit from maintaining high service standards to achieve and retain Superhost status.
- Positive guest experiences contribute to better ratings and repeat bookings.

6.8 Correlation Analysis
Correlation analysis identified relationships between price and several listing characteristics, including:
- Accommodation capacity
- Number of bedrooms
- Number of beds
- Review activity
- Occupancy-related measures

Business Interpretation
- Larger properties generally support higher pricing.
- Property characteristics influence perceived value and willingness to pay.
- Pricing decisions should consider both property size and location factors.

6.9 Key Findings Summary
The EDA revealed several important marketplace patterns:
- Entire-home listings command higher prices than private or shared rooms.
- Location is a major driver of price variation.
- Larger properties generally achieve higher prices.
- Listings with strong ratings and review histories perform better.
- Availability patterns provide useful indicators of demand and occupancy.
- Neighbourhood characteristics significantly influence market performance.

These findings provide the foundation for the statistical analysis and business recommendations presented in the following sections of the report.

7.Statistical Findings

Statistical analysis was conducted using **R** to evaluate key hypotheses related to Airbnb pricing, host performance, and neighbourhood characteristics. Hypothesis testing, effect size analysis, correlation analysis, regression modelling, and multicollinearity assessment were performed to support data-driven conclusions.

7.1 Hypothesis Testing Results

H1: Entire Home/Apt Listings vs. Private Rooms
A one-tailed independent samples t-test was conducted to compare listing prices between **Entire Home/Apt** and **Private Room** accommodations.

- The analysis indicated that Entire Home/Apt listings have significantly higher prices than Private Rooms.
- Cohen's *d* was calculated to measure the practical significance of the price difference.
- These findings suggest that guests are willing to pay a premium for greater privacy and exclusive use of the property.

H2: Superhost vs. Non-Superhost Ratings
An independent samples t-test compared review ratings between Superhosts and non-Superhosts.

- The results showed a difference in average review scores between the two groups.
- Cohen's *d* indicated the magnitude of this difference.
- This suggests that Superhost status is associated with higher guest satisfaction and service quality.

H3: Listings with More Than 10 Reviews vs. 10 or Fewer Reviews
A t-test was performed to determine whether review volume influences listing prices.

- Listings with more than ten reviews showed statistically different pricing compared to listings with ten or fewer reviews.
- Effect size analysis was used to evaluate the practical importance of the observed difference.
- This indicates that review activity may contribute to guest confidence and pricing strategies.

H4: Price Differences Across Neighbourhoods
A one-way Analysis of Variance (ANOVA) was conducted to compare average listing prices across neighbourhoods.

- The ANOVA results indicated statistically significant differences in average prices between neighbourhoods.
- Eta-squared (η^2) was calculated to measure the proportion of price variation explained by neighbourhood location.
- The findings confirm that location is one of the most influential factors affecting Airbnb pricing.

H5: Weekend vs. Weekday Pricing
This hypothesis could not be tested because the available **Boston calendar dataset did not contain a daily price variable**. The dataset included daily availability information but did not provide nightly prices required for comparing weekend and weekday pricing.

7.2 Effect Sizes
Effect sizes were reported alongside statistical significance tests to evaluate the practical importance of the findings.

- **Cohen's d** was used for independent samples t-tests (H1–H3).
- **Eta-squared (η^2)** was used for the ANOVA analysis (H4).

Reporting effect sizes provides additional insight into the magnitude of observed differences beyond statistical significance alone.

7.3 Correlation Analysis
Correlation analysis was conducted to identify relationships between numerical variables.
The analysis indicated positive relationships between listing price and several property characteristics, including:

- Accommodation capacity
- Number of bedrooms
- Number of beds

These variables were identified as important factors influencing Airbnb pricing.

7.4 Regression Analysis
Multiple Linear Regression (OLS) was used to identify the factors that significantly influence Airbnb listing prices.
The regression model included:

- Accommodation capacity
- Number of bedrooms
- Number of beds
- Room type
- Review score
- Occupancy rate
- Total number of reviews

The results showed that property characteristics and room type are among the strongest predictors of listing prices.

7.5 Multicollinearity Assessment
Variance Inflation Factor (VIF) analysis was performed to evaluate multicollinearity among the predictor variables.
The VIF values were within acceptable limits, indicating that multicollinearity was not a significant issue.
Therefore, the regression model provides reliable coefficient estimates and meaningful interpretation of the predictor variables.

7.6 Business Conclusions
The statistical analysis provides several important business insights:

- Entire-home listings command higher prices than private rooms, reflecting guests' preference for privacy and additional space.
- Superhosts generally receive higher customer ratings, highlighting the value of maintaining high service standards.
- Listings with greater review activity tend to demonstrate different pricing patterns, suggesting that guest feedback influences perceived value.
- Neighbourhood location has a significant impact on pricing, emphasizing the importance of location-based pricing strategies.

- Property size, room type, and accommodation capacity are key drivers of Airbnb listing prices.

Overall, the statistical findings support the use of data-driven pricing and host management strategies to improve competitiveness and maximize revenue in the Boston Airbnb marketplace.

8. Data Science Experiments

The objective of the data science experiments was to identify the key factors influencing Airbnb listing prices and evaluate the predictive capability of a statistical regression model. A Multiple Linear Regression (Ordinary Least Squares) model was developed using R to quantify the relationship between listing price and selected property characteristics.

8.1 Model Development

A Multiple Linear Regression (OLS) model was developed using the following predictor variables:

- Accommodation capacity
- Number of bedrooms
- Number of beds
- Room type
- Review score
- Occupancy rate
- Total number of reviews

The target variable was:

- Listing price

The model was selected because it provides interpretable coefficients, supports statistical inference, and is appropriate for examining the influence of multiple explanatory variables on a continuous outcome.

8.2 Model Validation

The regression model was evaluated using standard statistical measures, including:

- Coefficient estimates
- p-values for predictor significance
- R-squared and Adjusted R-squared
- Residual analysis
- Variance Inflation Factor (VIF)

These metrics were used to assess model performance and ensure that the assumptions of linear regression were reasonably satisfied.

8.3 Model Results

The regression analysis identified several significant predictors of Airbnb listing price.
Key findings include:

- Larger accommodation capacity was associated with higher listing prices.
- Listings with more bedrooms and beds generally commanded higher prices.
- Room type had a significant impact on pricing, with entire-home listings achieving higher prices than private or shared rooms.
- Review scores showed a positive relationship with listing price.
- Occupancy rate and review activity also contributed to explaining price variation.

8.4 Validation Metrics

The regression model was validated using the following metrics:

- **R-squared:** Measures the proportion of variation in listing price explained by the model.
- **Adjusted R-squared:** Evaluates explanatory power while accounting for the number of predictors.
- **p-values:** Identify statistically significant predictors.
- **Variance Inflation Factor (VIF):** Confirms that multicollinearity among predictors is within acceptable limits.

The VIF analysis indicated that multicollinearity was not a significant concern, supporting the reliability of the estimated regression coefficients.

8.5 Model Interpretation

The experimental results demonstrate that Airbnb pricing is influenced by a combination of property characteristics, room type, and customer-related factors.

The strongest contributors to higher listing prices were:

- Accommodation capacity
- Number of bedrooms
- Room type
- Property size

Review-related variables contributed positively but had a comparatively smaller influence on price.

8.6 Business Implications

The regression model provides practical insights for Airbnb hosts and property managers.

Key recommendations include:

- Price listings according to accommodation capacity and property size.
- Maintain high-quality properties and guest experiences to improve review scores.
- Use neighbourhood and room type information when developing pricing strategies.
- Monitor occupancy and review activity to optimize pricing decisions.

Overall, the regression model provides a statistically supported framework for understanding the factors that drive Airbnb listing prices and supports evidence-based decision-making.

9.AI/ML Experiments

9.1 AI/ML Experiment Overview

The primary focus of this project was data engineering, statistical analysis, and business intelligence. An AI/ML experiment **was not implemented** as part of the current analysis because the selected Airbnb dataset mainly contains structured numerical and categorical information, with limited unstructured text data suitable for advanced NLP or Large Language Model (LLM) applications.

However, potential AI/ML approaches were evaluated to identify future opportunities for improving Airbnb marketplace analysis.

9.2 Potential NLP Experiment: Review Sentiment Analysis

Customer reviews contain valuable information about guest experiences and service quality. A potential NLP-based experiment could analyze review comments to classify customer sentiment.

Methodology
The proposed approach would include:
- Extracting review text from the reviews dataset.
- Cleaning text data by removing unnecessary characters, stop words, and applying text normalization.
- Applying sentiment analysis techniques to classify reviews as:
  - Positive
  - Neutral
  - Negative
- Generating sentiment scores and combining them with listing information.

Possible techniques:
- TF-IDF feature extraction
- Logistic Regression classification
- Transformer-based models such as BERT

9.3 Expected Results and Business Value

A sentiment analysis model could provide insights such as:
- Identifying common customer complaints.
- Understanding factors contributing to positive experiences.
- Comparing sentiment scores across neighbourhoods and property types.
- Helping hosts improve service quality.

For example:
- Negative sentiment related to cleanliness may indicate areas requiring operational improvements.
- Positive sentiment related to location or amenities may highlight competitive advantages.

9.4 Critical Evaluation
Although NLP-based sentiment analysis could provide additional insights, there are several limitations:
- Review text availability varies between listings.
- Some listings have limited customer feedback.

- Sentiment models may struggle with sarcasm, informal language, and context-specific expressions.
- Additional preprocessing and model validation would be required before production use.

9.5 Future AI/ML Opportunities

Future extensions of this project could include:

- **Price Prediction Model:**
  Machine learning models such as Random Forest, XGBoost, or Gradient Boosting could be used to predict optimal listing prices.
- **Recommendation System:**
  A recommendation engine could suggest suitable Airbnb listings based on user preferences, location, budget, and previous behaviour.
- **LLM-based Business Assistant:**
  A large language model application could provide automated insights from Airbnb data, such as answering business questions about pricing trends and neighbourhood performance.
- **Anomaly Detection:**
  Machine learning techniques could identify unusual pricing patterns, suspicious listings, or abnormal availability behaviour.

9.6 Conclusion

Although advanced AI/ML models were not implemented in this project, the existing data pipeline and analytical foundation provide a strong base for future AI-driven applications. The integration of NLP, predictive modelling, and LLM-based analytics could further enhance decision-making capabilities in the Airbnb marketplace.

10. Visualizations

Interactive and analytical visualizations were developed using **Power BI and R** to communicate key findings from the Airbnb dataset. The visualizations were designed to support business understanding of listing performance, pricing patterns, host behaviour, and marketplace characteristics.

Figure 1: Airbnb Performance Overview Dashboard

Visualization Type: Power BI Dashboard

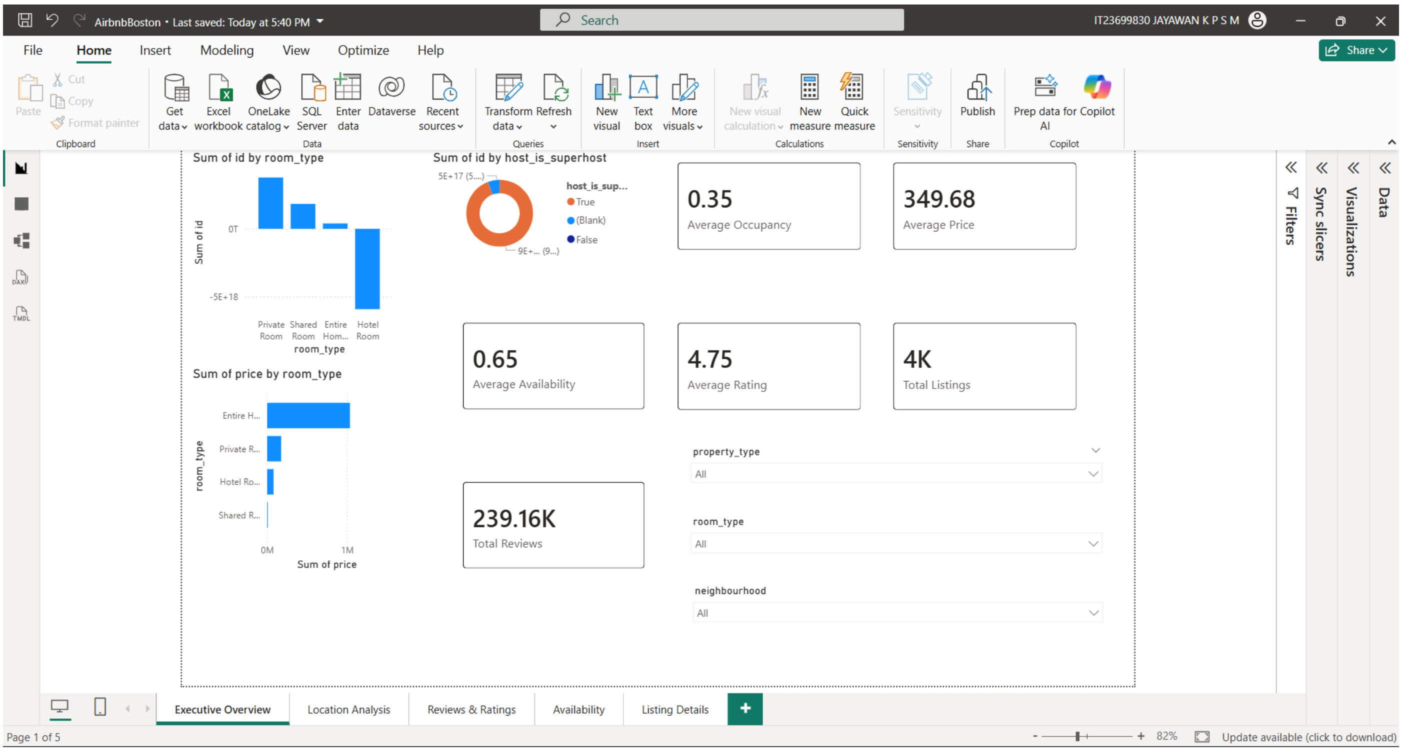

Figure 2 - Airbnb Performance Overview Dashboard

Description:
A summary dashboard was created to provide an overview of Airbnb marketplace performance using key performance indicators (KPIs).

Key Elements:

- Total number of listings
- Average listing price
- Host performance indicators
- Review-related metrics

Interpretation:
The dashboard provides stakeholders with a high-level view of the Airbnb marketplace and allows quick assessment of overall listing activity and performance.

Business Insight:
Decision-makers can use these indicators to understand marketplace size, pricing trends, and host performance.

Figure 2: Listing Distribution Analysis Using Slicers

Visualization Type: Power BI Slicer Analysis

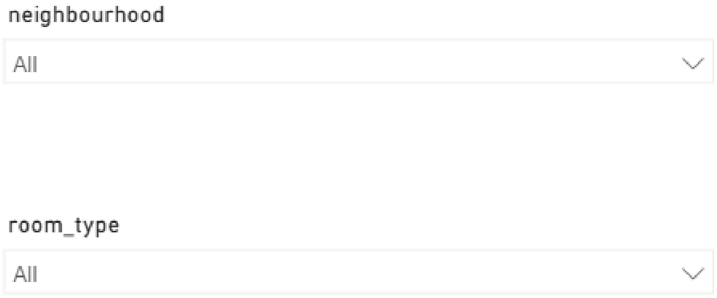

Figure 3.1 Figure 2 - Interactive Listing Filtering Using Slicers

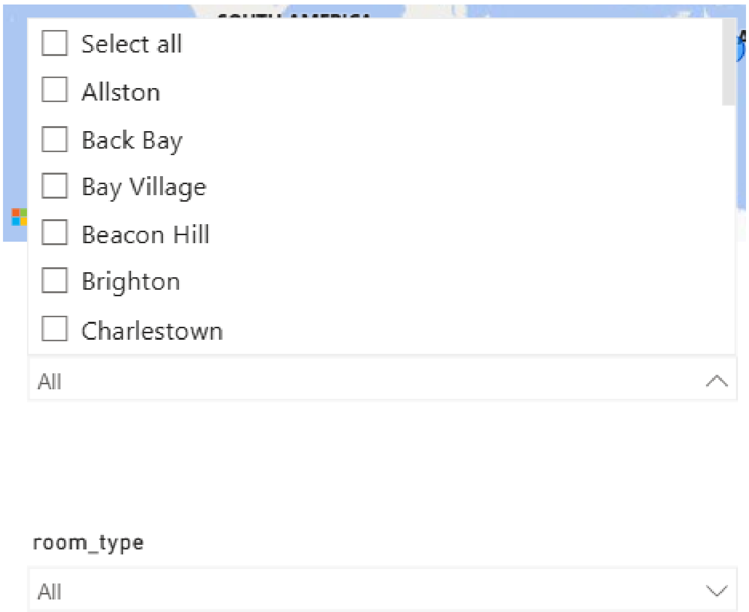

Figure 3.2

Description:
Slicers were implemented to allow users to filter Airbnb listings dynamically based on different attributes.

Filters Included:
- Neighbourhood
- Room type

- Property characteristics

Interpretation:
Interactive filtering allows users to explore how listing characteristics change across different market segments.

Business Insight:
Hosts and analysts can compare similar properties and identify differences between locations and accommodation types.

Figure 3: Room Type Pricing Analysis

Visualization Type: Power BI Chart

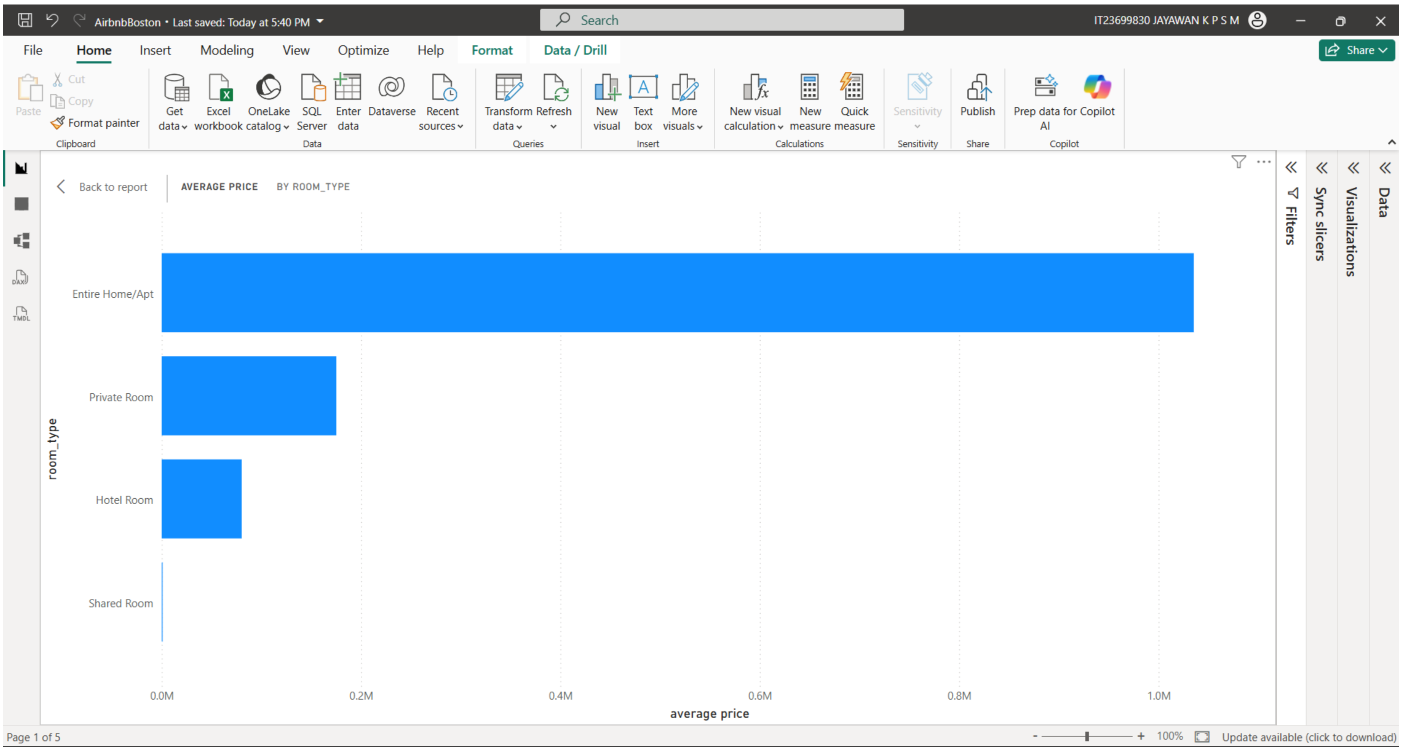

Figure 4 - Average Price Comparison by Room Type

Description:
The visualization compares pricing differences between different room categories.

Interpretation:
Entire-home listings generally show higher prices compared with private and shared rooms.

Business Insight:
Property type significantly influences customer willingness to pay, and hosts can adjust pricing strategies based on accommodation type.

Figure 4: Neighbourhood Price Comparison

Visualization Type: Power BI Visual

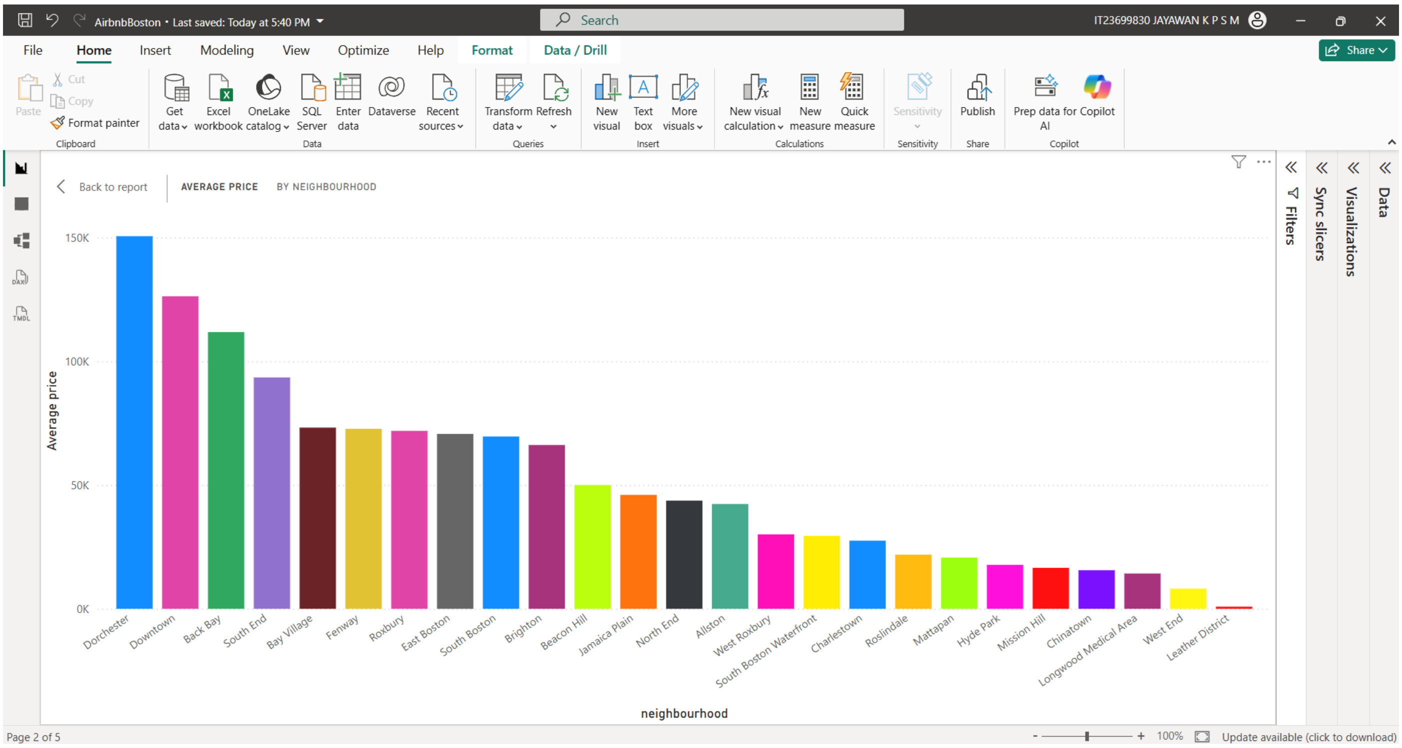

Figure 5.1 - Price Variation Across Neighbourhoods

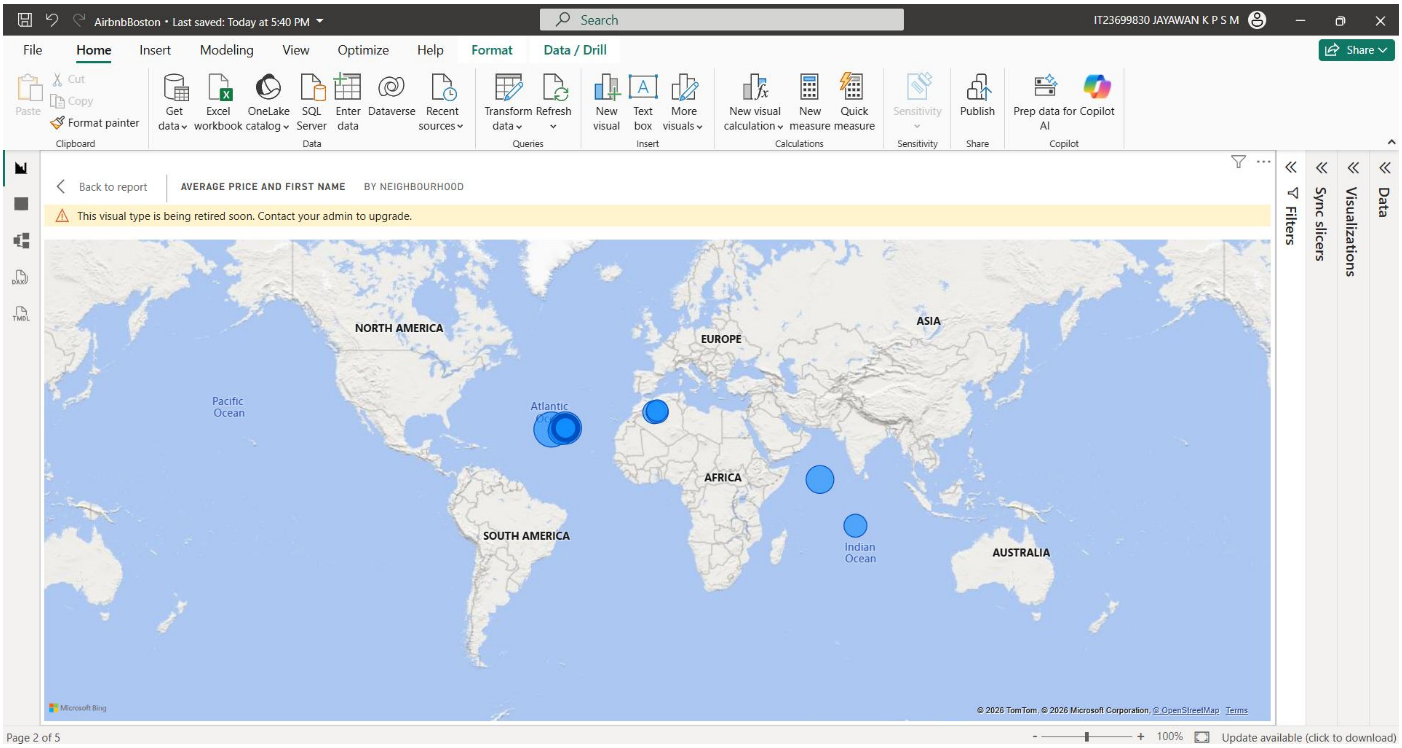

Figure 5.2 - Price Variation Across Neighbourhoods

Description:
This visualization compares Airbnb performance across different neighbourhoods.

Interpretation:
Pricing varies across neighbourhoods, showing that location has an important impact on Airbnb market value.
Business Insight:
Hosts should consider neighbourhood-specific pricing rather than applying the same pricing strategy across all locations.

Figure 5: Matrix Report for Detailed Data Analysis

Visualization Type: Power BI Matrix Visual

Sum of price	host_name	Sum of availability_rate	Sum of occupancy_rate	name
235.84	Katarzyna	0.61	0.39	"Private Guest Floor in Single-Family Home"
485.00	Foster Love	0.44	0.56	(Southie1) Bright 2BR Stay Near Seaport & Downtown
371.00	Foster Love	0.45	0.55	(Southie2) Modern 2BR –Walk to Shops & Dining, W/D
482.50	Foster Love	0.41	0.59	(Southie3) South Boston Gem w/ AC + LaundryW/D
311.00	Foster Love	0.45	0.55	(Southie4) Sunny 2BR Near Seaport & Downtown
285.33	Sandy	0.59	0.41	1 BR Lux Apt, Wi-Fi, View of Boston, Free Parking
605.96	Lucio	0.59	0.41	2 BDR/2b- walk to Seaport & BCEC
136.62	Evolve	0.53	0.47	2 Mi to Dtwn: Boston Apartment Near Beaches!
434.00	Elissa	0.01	0.99	2BR walking distance to seaport/downtown
660.00	Bennett	0.08	0.92	3BR | 3.5BA Single Family | Workspace | Playroom
709.00	Maureen	0.68	0.32	4br SouthBoston free parkg nrBeach, 6 min>BCEC
528.16	Jessica	0.48	0.52	5BR 3BATH LUX | Walk to Subway
973.86	Jessica	1.00	0.00	5BR 3Bth | Redline Subway Access
1,758.00	Jessica	0.79	0.21	5BR3B By Subway Downtown
132.96	Pei	0.79	0.21	A Comfortable place close to Boston downtown
289.97	Rachel	0.65	0.35	A peaceful home in the heart of South Boston
638.28	Lindsay	1.00	0.00	Adorable private suite with a private entrance and bathroom close to Downtown, Seaport District and The BCE
109.40	Elizabeth	0.23	0.77	Beautiful 1 Bedroom Condo in Heart of South Boston
884.67	Ty	0.21	0.79	Beautiful Oceanfront, 2Bed/Bath South Boston condo
814.33	Charles	0.55	0.45	Beautiful Southie Single Family
69,628.13		**83.74**	**62.26**	

Figure 6 - Airbnb Listing Matrix Analysis

Description:
A matrix report was created to display detailed Airbnb information using row and column groupings.

Features:

- Hierarchical data organization
- Detailed comparison between categories
- Structured tabular analysis

Interpretation:
The matrix allows users to analyze detailed relationships between listing attributes and performance measures.

Business Insight:
Users can identify patterns across different property segments and neighbourhood categories.

Figure 6: Drill-down Analysis Report

Visualization Type: Power BI Drill-down Report

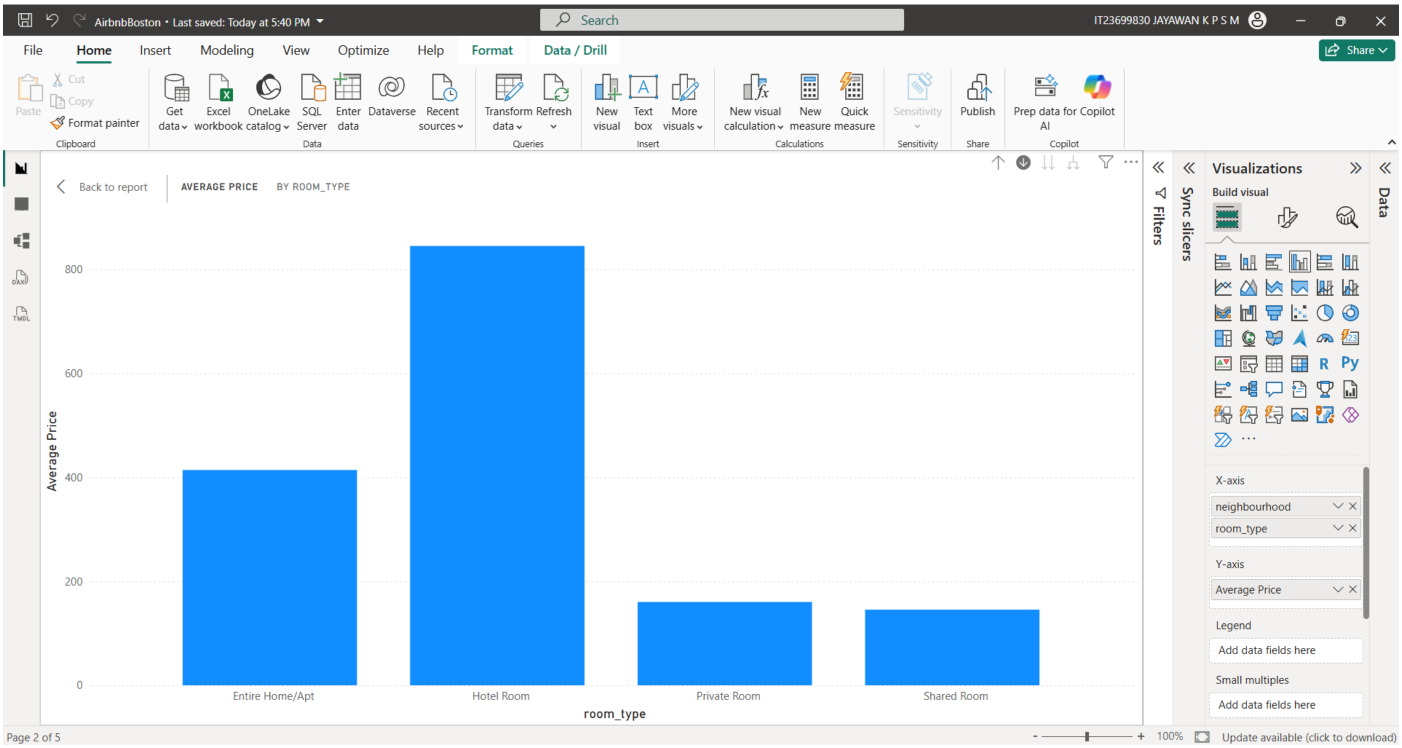

Figure 7.1 - Airbnb Listing Matrix Analysis

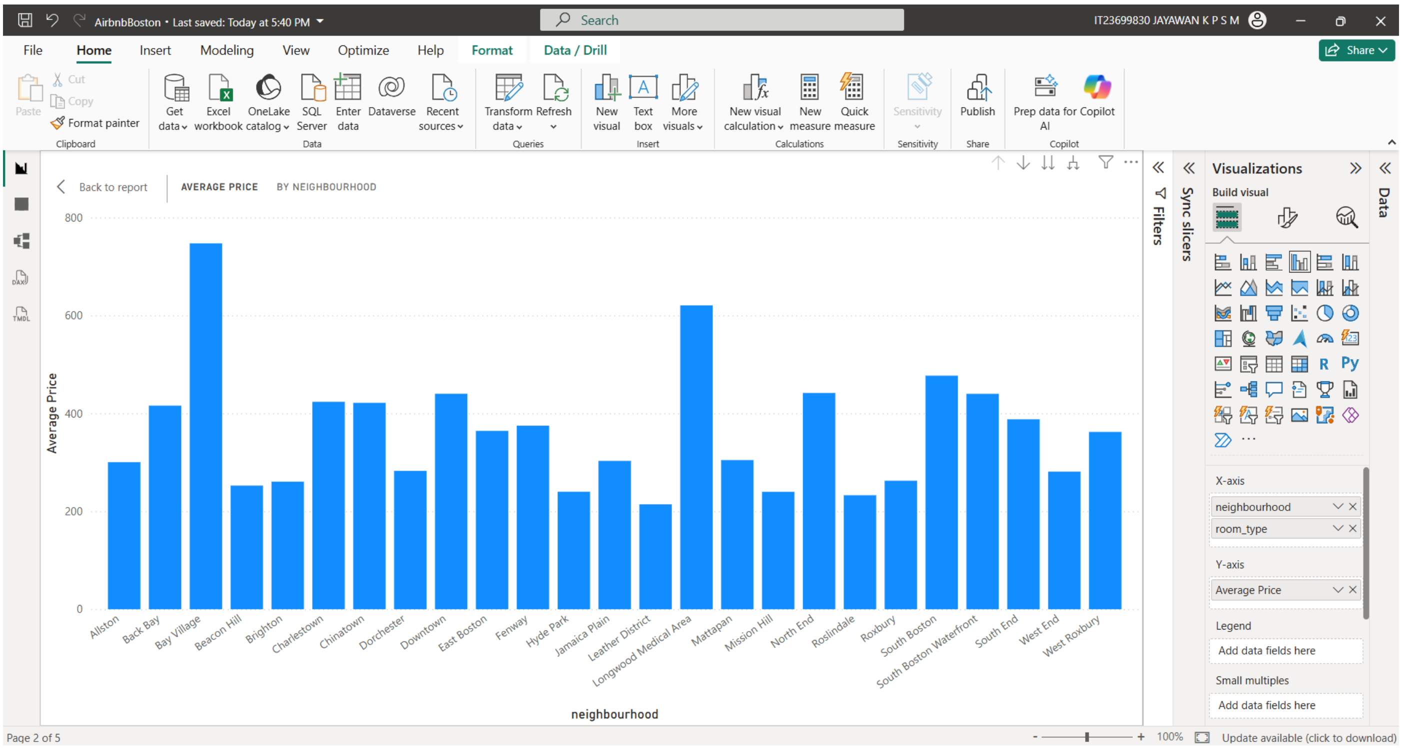

Figure 7.2 - Airbnb Listing Matrix Analysis

Description:
A drill-down report was developed to allow users to explore information hierarchically.

Example Hierarchy:
Neighbourhood → Listing Category → Detailed Metrics

Interpretation:
Users can move from summarized information to detailed levels of analysis.

Business Insight:
Drill-down functionality helps stakeholders investigate specific areas of interest without viewing unnecessary details.

Figure 7: Drill-through Detailed Listing Analysis

Visualization Type: Power BI Drill-through Report

Sum of price	host_name	Sum of availability_rate	Sum of occupancy_rate	name
235.84	Katarzyna	0.61	0.39	"Private Guest Floor in Single-Family Home"
485.00	Foster Love	0.44	0.56	(Southie1) Bright 2BR Stay Near Seaport & Downtown
371.00	Foster Love	0.45	0.55	(Southie2) Modern 2BR –Walk to Shops & Dining, W/D
482.50	Foster Love	0.41	0.59	(Southie3) South Boston Gem w/ AC + LaundryW/D
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285.33	Sandy	0.59	0.41	1 BR Lux Apt, Wi-Fi, View of Boston, Free Parking
605.96	Lucio	0.59	0.41	2 BDR/2b- walk to Seaport & BCEC
136.62	Evolve	0.53	0.47	2 Mi to Dtwn: Boston Apartment Near Beaches!
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660.00	Bennett	0.08	0.92	3BR | 3.5BA Single Family | Workspace | Playroom
709.00	Maureen	0.68	0.32	4br SouthBoston free parkg nrBeach, 6 min>BCEC
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814.33	Charles	0.55	0.45	Beautiful Southie Single Family
69,628.13		83.74	62.26	

Figure 8 - Drill-through Listing Detail Report

Description:
A drill-through page was created to provide detailed information about selected listings.

Features:
- Detailed listing information
- Related performance metrics
- Individual record analysis

Interpretation:
Users can right-click on a visual element and navigate to a detailed analysis page.

Business Insight:
This supports deeper investigation of individual listings and helps identify factors affecting performance.

Figure 8: Correlation Analysis Visualization

Visualization Type: R Statistical Visualization

```
                           price accommodates    bedrooms         beds review_scores_rating
price                1.000000000  0.605486166  0.55754179  0.542253055           0.11041385
accommodates         0.605486166  1.000000000  0.84141249  0.851722646           0.09492577
bedrooms             0.557541789  0.841412491  1.00000000  0.794342667           0.09810432
beds                 0.542253055  0.851722646  0.79434267  1.000000000           0.06421436
review_scores_rating 0.110413852  0.094925770  0.09810432  0.064214356           1.00000000
occupancy_rate       0.040298170 -0.007085351  0.03187907 -0.023886873           0.08120989
total_reviews       -0.005384338 -0.032904002 -0.05092970 -0.009423277           0.09138434
availability_30      0.093930126  0.031512256 -0.01266376  0.035260168          -0.14634378
availability_60      0.070126362  0.014195524 -0.02452765  0.022314988          -0.15754723
availability_90      0.056490619  0.007411749 -0.03285096  0.022615783          -0.15399445
availability_365    -0.040298170  0.007085351 -0.03187907  0.023886873          -0.08120989
                    occupancy_rate total_reviews availability_30 availability_60 availability_90
price                  0.040298170  -0.005384338      0.09393013      0.07012636     0.056490619
accommodates          -0.007085351  -0.032904002      0.03151226      0.01419552     0.007411749
bedrooms               0.031879075  -0.050929702     -0.01266376     -0.02452765    -0.032850956
beds                  -0.023886873  -0.009423277      0.03526017      0.02231499     0.022615783
review_scores_rating   0.081209885   0.091384342     -0.14634378     -0.15754723    -0.153994453
occupancy_rate         1.000000000   0.174252167     -0.33490827     -0.39978733    -0.477991001
total_reviews          0.174252167   1.000000000     -0.13287187     -0.13048079    -0.143293635
availability_30       -0.334908270  -0.132871866      1.00000000      0.90699324     0.822799134
availability_60       -0.399787328  -0.130480792      0.90699324      1.00000000     0.950105942
availability_90       -0.477991001  -0.143293635      0.82279913      0.95010594     1.000000000
availability_365      -1.000000000  -0.174252167      0.33490827      0.39978733     0.477991001
                    availability_365
price                   -0.040298170
accommodates             0.007085351
bedrooms                -0.031879075
beds                     0.023886873
review_scores_rating    -0.081209885
occupancy_rate          -1.000000000
total_reviews           -0.174252167
availability_30          0.334908270
availability_60          0.399787328
availability_90          0.477991001
availability_365         1.000000000
```

Figure 9 - Correlation Analysis of Numerical Variables

Description:
Correlation analysis was performed to identify relationships between numerical variables.

Variables Analyzed Included:

- Price
- Accommodation capacity
- Bedrooms
- Beds
- Review metrics
- Availability measures

Interpretation:
The analysis identified relationships between property characteristics and listing prices.

Business Insight:
Property size-related factors are important considerations when developing pricing strategies.

Figure 9: Regression Model Results Visualization

Visualization Type: R Statistical Output

```
Call:
lm(formula = price ~ accommodates + bedrooms + beds + review_scores_rating +
    occupancy_rate + total_reviews + room_type, data = regression_data)

Residuals:
   Min     1Q Median     3Q    Max
-772.5 -106.1  -26.5   67.1 3305.1

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)           -1.217e+02  6.029e+01  -2.018 0.043663 *
accommodates           3.647e+01  3.716e+00   9.815  < 2e-16 ***
bedrooms               4.960e+01  7.833e+00   6.332 2.77e-10 ***
beds                   1.494e+01  5.024e+00   2.974 0.002965 **
review_scores_rating   4.814e+01  1.264e+01   3.807 0.000143 ***
occupancy_rate         5.253e+01  1.588e+01   3.308 0.000949 ***
total_reviews         -9.036e-03  3.904e-02  -0.231 0.816971
room_typeHotel Room    3.634e+02  4.009e+01   9.065  < 2e-16 ***
room_typePrivate Room -1.085e+02  1.046e+01 -10.370  < 2e-16 ***
room_typeShared Room  -7.036e+01  1.375e+02  -0.512 0.608768
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 237.6 on 3067 degrees of freedom
Multiple R-squared:  0.4177,	Adjusted R-squared:  0.416
F-statistic: 244.5 on 9 and 3067 DF,  p-value: < 2.2e-16
```

Figure 10 - Multiple Linear Regression Results

Description:
Regression analysis results were used to evaluate factors influencing Airbnb listing prices.

Analyzed Factors Included:

- Accommodation capacity
- Bedrooms
- Beds
- Room type
- Review scores
- Occupancy rate
- Total reviews

Interpretation:
The model results highlight the variables that contribute most to explaining price variation.

Business Insight:
Hosts can use these factors to make informed pricing decisions.

Figure 10: Statistical Analysis Results Summary

Visualization Type: R Analysis Outputs

```
        Welch Two Sample t-test

data:  price by room_type
t = 31.606, df = 3562.7, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Entire Home/Apt and group Private Room is
greater than 0
95 percent confidence interval:
 240.5979      Inf
sample estimates:
mean in group Entire Home/Apt    mean in group Private Room
                     414.0002                      160.1902
```

```
        Welch Two Sample t-test

data:  review_scores_rating by host_is_superhost
t = -16.41, df = 2654.7, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 -0.2074644 -0.1631755
sample estimates:
 mean in group No mean in group Yes
         4.667791          4.853111
```

```
        Welch Two Sample t-test

data:  price by review_group
t = -3.0801, df = 1479.7, p-value = 0.002107
alternative hypothesis: true difference in means between group 10 or fewer reviews and group More than 1
0 reviews is not equal to 0
95 percent confidence interval:
 -67.10925 -14.88890
sample estimates:
 mean in group 10 or fewer reviews mean in group More than 10 reviews
                          312.1092                           353.1083
```

```
> summary(anova_model)
                Df    Sum Sq Mean Sq F value Pr(>F)
neighbourhood   24  37906941 1579456   15.88 <2e-16 ***
Residuals     3668 364761288   99444
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> eta_squared(anova_model)
For one-way between subjects designs, partial eta squared is equivalent to eta squared.
  Returning eta squared.
# Effect Size for ANOVA

Parameter     | Eta2 |       95% CI
-----------------------------------
neighbourhood | 0.09 | [0.07, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

Figure 11 - Statistical Hypothesis Testing Results

Description:
Statistical testing outputs were used to support hypothesis evaluation.

Included Analyses:

- T-tests
- ANOVA results
- Effect size measurements

Interpretation:
The statistical results provide evidence for differences in pricing, host performance, and neighbourhood effects.

Business Insight:
Data-driven statistical evidence supports more reliable marketplace decisions compared with assumptions based only on observation.

The developed visualizations successfully communicate important Airbnb marketplace insights through interactive dashboards and statistical analysis outputs. Power BI enabled business-focused exploration, while R supported statistical validation of observed patterns. Together, these visualizations helped identify pricing drivers, neighbourhood differences, and performance factors affecting Airbnb listings.

11. Business Recommendations

Based on the exploratory analysis, statistical findings, regression analysis, and Power BI dashboard insights, the following recommendations are proposed to improve Airbnb listing performance and support data-driven decision-making.

1. Implement Data-Driven Pricing Strategies

Recommendation:
Hosts should adjust pricing based on property characteristics, room type, and neighbourhood demand.
Evidence from Analysis:
- Statistical testing showed significant price differences between room types.
- Regression analysis identified accommodation capacity, bedrooms, beds, and room type as important price predictors.

Business Impact:
Dynamic pricing strategies can help hosts maximize revenue while remaining competitive within their market segment.

2. Optimize Pricing Based on Property Characteristics

Recommendation:
Hosts should consider property size and capacity when setting listing prices.
Evidence from Analysis:
- Larger properties with higher accommodation capacity generally achieved higher prices.
- Bedrooms and beds showed positive relationships with listing prices.

Business Impact:
Understanding the value contribution of property features allows hosts to set more accurate and competitive prices.

3. Focus on Improving Guest Experience

Recommendation:
Hosts should maintain high service quality to improve guest satisfaction and strengthen marketplace performance.
Evidence from Analysis:
- Superhost analysis showed differences in review performance.
- Review scores were included as factors influencing listing performance.

Business Impact:
Higher guest satisfaction can improve reputation, increase booking confidence, and support long-term revenue growth.

4. Develop Neighbourhood-Specific Strategies

Recommendation:
Pricing and marketing strategies should be customized according to neighbourhood characteristics.
Evidence from Analysis:
- ANOVA results identified significant price differences between neighbourhoods.

- Power BI analysis showed variations in listing performance across locations.

Business Impact:
Location-based strategies allow hosts to better match pricing with local demand and competition levels.

5. Use Analytics Dashboards for Continuous Monitoring

Recommendation:
Stakeholders should regularly monitor Airbnb performance using interactive dashboards.
Evidence from Analysis:

- Power BI dashboards provided interactive exploration through KPI cards, slicers, matrix analysis, drill-down, and drill-through reports.

Business Impact:
Continuous monitoring enables faster identification of market changes and supports informed business decisions.

6. Improve Data Quality Management

Recommendation:
Future data pipelines should include stronger validation and monitoring processes.
Evidence from Data Engineering Analysis:

- Missing values, duplicate records, and inconsistent fields were identified during data cleaning.
- Data validation rules were applied during preprocessing.

Business Impact:
Improved data quality increases the reliability of future analytics and machine learning applications.

7. Future Predictive Analytics Implementation

Recommendation:
Future improvements should include machine learning models for price prediction and demand forecasting.
Evidence from Analysis:

- Regression analysis identified important pricing drivers.
- The existing data pipeline provides a foundation for predictive modelling.

Business Impact:
Predictive models could help automate pricing recommendations and improve revenue optimization.

Overall Recommendation
The Airbnb marketplace should adopt a data-driven decision-making approach by combining reliable data engineering practices, statistical analysis, and interactive business intelligence dashboards. Using insights from property characteristics, location, and customer feedback can help hosts optimize pricing, improve customer experience, and increase marketplace competitiveness.

12. Cross-City Comparisons

Although the Airbnb dataset provided multiple city datasets, this project focused only on the **Boston Airbnb market**. The decision to select a single city was made to allow a more detailed analysis of pricing behaviour, listing characteristics, host performance, neighbourhood patterns, and availability trends.

City Selection Rationale
Boston was selected because:

- It provides a sufficiently large number of Airbnb listings for meaningful statistical analysis.
- The dataset contains complete information across listings, reviews, calendar availability, and neighbourhood attributes.
- Focusing on one city allowed deeper exploration and more detailed business insights within a specific marketplace.

Limitation
Cross-city comparisons were not performed because only one city dataset was processed and analysed. Therefore, regional pricing differences, market dynamics, and variations between cities could not be evaluated.

Future Enhancement
A multi-city analysis could be implemented by applying the same data pipeline and analytical framework to additional cities. Future comparisons could investigate:

- Differences in average listing prices across cities.
- Variations in room-type popularity.
- Regional demand and availability patterns.
- Differences in host performance and review behaviour.
- Location-based pricing strategies across markets.

13. Limitations & Caveats

This project provides valuable insights into the Boston Airbnb marketplace; however, several limitations and assumptions should be considered when interpreting the results.

13.1 Dataset Availability and Scope Limitations

- The original assignment provided multiple city datasets; however, only the **Boston dataset was available for analysis during this project**. Therefore, the analysis was limited to a single-city case study.
- Due to the lack of additional city datasets, cross-city comparisons and regional market comparisons could not be performed.
- The findings and recommendations are specific to the Boston Airbnb market and may not directly apply to other cities with different customer behaviours, regulations, and market conditions.

13.2 Data Quality Limitations

- The dataset contained missing values in some attributes, particularly review-related and host information fields.
- Some data cleaning and standardization were required before analysis, including handling missing values, formatting inconsistencies, and duplicate records.
- Price-related variables may contain extreme values from luxury listings, which can influence average price calculations.

13.3 Statistical Analysis Limitations

- The statistical analysis was based only on the variables available in the Boston dataset.
- Some external factors affecting Airbnb prices, such as seasonal events, tourism trends, economic conditions, and local regulations, were not included.
- Regression analysis identifies relationships between variables but does not prove direct causal relationships.
- The regression model assumes linear relationships and may not capture complex non-linear pricing patterns.

13.4 Calendar Data Limitation

- The available calendar dataset contained availability and booking-related information but did not include daily pricing information.
- Therefore, the hypothesis testing for **weekend versus weekday price differences (H5)** could not be performed.

13.5 Model and Scope Limitations

- The project focused on exploratory analysis, statistical testing, regression modelling, and business intelligence reporting.
- Advanced machine learning approaches such as price prediction models, recommendation systems, NLP sentiment analysis, and LLM-based applications were not implemented.
- The developed insights are based on historical Airbnb data and may require future updates when market conditions change.

Conclusion

Despite these limitations, the project demonstrates a complete analytical workflow using data engineering, statistical analysis, and visualization techniques. The Boston dataset provided sufficient information to identify key pricing drivers, neighbourhood patterns, and host performance factors while highlighting opportunities for future multi-city and AI-driven extensions.

14. Future Improvements

With additional time, more comprehensive datasets, and additional resources, this project could be extended to provide deeper insights and more advanced analytical capabilities.

14.1 Multi-City Market Analysis

- Include additional city datasets to perform cross-city comparisons.
- Compare pricing patterns, customer behaviour, and availability trends across different markets.
- Identify regional differences in Airbnb demand and host strategies.

Expected Benefit:
A broader understanding of Airbnb marketplace dynamics across different locations.

14.2 Advanced Machine Learning Models

- Develop predictive models for Airbnb price estimation using machine learning algorithms such as:
    - Random Forest
    - XGBoost
    - Gradient Boosting
- Compare model performance using evaluation metrics such as:
    - RMSE
    - MAE
    - R^2 score

Expected Benefit:
More accurate price prediction and automated pricing recommendations for hosts.

14.3 Real-Time Data Pipeline Implementation

- Develop an automated pipeline to regularly collect and process updated Airbnb data.
- Implement scheduled data ingestion and incremental loading.
- Add monitoring, logging, and automated error handling.

Expected Benefit:
A production-ready analytics system capable of handling continuously changing marketplace data.

14.4 Customer Review Sentiment Analysis

- Integrate review text data and apply Natural Language Processing (NLP) techniques.
- Classify reviews into positive, neutral, and negative sentiments.
- Identify common factors affecting guest satisfaction.

Expected Benefit:
Better understanding of customer experiences and areas for service improvement.

14.5 Enhanced Feature Engineering

Future analysis could include additional derived features such as:

- Price per guest
- Price per bedroom
- Seasonal demand indicators
- Distance from popular locations
- Amenity-based scoring
- Booking trend features

Expected Benefit:
Improved understanding of factors influencing listing performance.

14.6 Improved Statistical and Predictive Modelling

- Apply advanced statistical techniques to capture non-linear relationships.
- Perform time-series analysis using calendar data when daily pricing information is available.
- Apply additional validation methods to improve model reliability.

Expected Benefit:
More robust insights and stronger predictive capabilities.

14.7 Dashboard Enhancement

Future Power BI improvements could include:

- Real-time data refresh
- More interactive filtering options
- Additional geographic visualizations
- Automated KPI monitoring
- Predictive analytics integration

Expected Benefit:
Improved decision support for hosts and business stakeholders.

Conclusion

Future improvements would focus on expanding the dataset coverage, introducing advanced AI/ML techniques, automating the data pipeline, and improving analytical capabilities. These enhancements would transform the current analytical solution into a more scalable and intelligent Airbnb marketplace decision-support system.

15. Reflection

This project provided practical experience in applying the complete data analytics workflow, from data preparation and modelling to statistical analysis and business reporting. Throughout the project, decisions were made to balance project scope, available resources, and analytical depth.

15.1 Work Prioritization

The project activities were prioritized based on their importance for delivering meaningful business insights. The main priorities were:

- Preparing and cleaning the Airbnb dataset to ensure data quality before analysis.
- Building a structured master dataset by integrating listing, review, and calendar information.
- Performing exploratory data analysis to identify important marketplace patterns.
- Applying statistical methods to validate business assumptions.
- Developing Power BI dashboards to communicate findings effectively.

Due to limited time and dataset availability, the focus was placed on completing a high-quality analysis of the Boston Airbnb dataset rather than expanding the project scope.

15.2 Trade-offs Made

Several trade-offs were considered during the project:

- **Single-city analysis vs. multi-city analysis:**
  Although multiple city datasets were provided, the analysis was limited to Boston because only this dataset was available and focusing on one city allowed deeper exploration of pricing, neighbourhood, and host behaviour.
- **Advanced machine learning vs. statistical modelling:**
  Instead of implementing complex machine learning models, the project focused on regression analysis and hypothesis testing because these methods provide clear explanations of factors influencing Airbnb prices.
- **Completeness vs. time constraints:**
  Some advanced features, such as NLP-based review analysis, real-time pipelines, and predictive modelling, were identified as future improvements rather than implemented within the project timeframe.
- **Data availability limitations:**
  The weekend versus weekday pricing hypothesis could not be evaluated because the available calendar dataset did not contain daily price information.

15.3 Key Lessons Learned

Through this project, several important lessons were gained:

- **Data quality is essential:**
  Reliable analysis depends on proper cleaning, validation, and handling of missing values before modelling.
- **Business understanding is important:**
  Statistical results become more valuable when translated into practical business recommendations.
- **Different tools serve different purposes:**
  Python was useful for data processing, R supported statistical analysis, and Power BI helped communicate insights through interactive reporting.
- **Model interpretation matters:**
  Understanding why variables influence outcomes is as important as achieving statistical results.
- **Scope management is critical:**
  Defining realistic project boundaries helps deliver meaningful results within available time and resources.

Conclusion

This project strengthened my understanding of the end-to-end data analytics process, including data engineering, statistical analysis, visualization, and business interpretation. The experience highlighted the importance of making informed trade-offs, documenting limitations, and selecting appropriate analytical methods based on available data and project objectives.

16. Appendix

AI tools were used as a supporting resource during this project to assist with technical guidance, documentation improvement, and problem-solving. All analysis decisions, code execution, data processing, and interpretation of results were reviewed and validated by the project author.

A.1 AI Tools Used

Tool Used:
- ChatGPT (OpenAI)

Purpose of Usage:
- Assistance with understanding data engineering and analytics concepts.
- Debugging programming errors in Python, R, SQL, and Power BI workflows.
- Improving report structure and technical documentation.
- Providing guidance on statistical analysis methods and interpretation.

A.2 Assisted Sections

AI assistance was used in the following areas:

Data Engineering Support
AI was used to assist with:
- Understanding data cleaning approaches.
- Troubleshooting SQL import errors.
- Suggesting solutions for duplicate records, missing values, and data type issues.
- Providing guidance on creating a structured analytical dataset.

Statistical Analysis Support
AI assisted with:
- Explaining hypothesis testing methods.
- Troubleshooting R errors related to:
    - t-tests
    - ANOVA
    - Cohen's d effect size
    - Regression modelling
    - VIF analysis
- Helping interpret statistical outputs in business terms.

Power BI Documentation Support
AI assisted with:
- Explaining how to create:
    - KPI cards
    - Slicers
    - Matrix visuals
    - Drill-down reports
    - Drill-through reports
- Structuring dashboard descriptions for documentation.

Report Writing Support
AI assisted with:

- Improving the structure and clarity of report sections.
- Drafting explanations for:
  - Methodology
  - Statistical findings
  - Business recommendations
  - Limitations
  - Future improvements
  - Reflection

A.3 Example Prompts Used

Examples of prompts used during the project include:

- "Explain this SQL import error and how to fix it."
- "Help me clean Airbnb dataset columns using Python."
- "How can I perform hypothesis testing for Airbnb pricing in R?"
- "Explain this regression output in business terms."
- "Help me write the statistical findings section for my report."
- "How can I create a Matrix visual and drill-through report in Power BI?"
- "Review this project section and improve the documentation."

A.4 Validation and Verification Process

AI-generated suggestions were not directly accepted without verification.
Validation steps included:

- Running all Python and R code locally and checking outputs.
- Confirming statistical results using R output.
- Checking Power BI visuals manually.
- Reviewing dataset columns before applying transformations.
- Comparing AI suggestions with assignment requirements.
- Ensuring that only completed project activities were included in the final documentation.

A.5 Limitations of AI Assistance

While AI provided useful technical support, it was not used to replace analytical decision-making.
The following tasks were completed independently:

- Dataset selection and project scope decisions.
- Data cleaning execution.
- Database implementation.
- Statistical analysis execution.
- Dashboard development.
- Interpretation of project-specific findings.

AI was used only as a supporting tool for learning, debugging, and improving documentation quality.